

Quantum Mechanics — Combined Short Notes

Part I: Balki sir, Lec-6 to Lec-13 · Part II: Short notes (Q.M), Pages 1–10

Contents

Part I — Short Notes (Q.M) [Balki sir], Lec-6 to Lec-13 1 (Short Notes) (Q.M) [Balki sir] Lec-6 1 The Schrödinger equation :- 1 (Short Notes) (Q.M) [Balki sir] Lec-7,8 2 (Short Notes) (Q.M) [Balki sir] Lec-8, 9 3 (Short Notes) (Q.M) [Balki sir] Lec-9,10 4 Short notes (Balki sir) Lec-11 5 Short notes (Balki sir) Lec-11 7 Short notes (Balki sir) Lec-11,12 8 Short notes (Balki sir) Lec-12,13 10 Short notes (Balki sir) Lec-12,13 11 Short notes (Balki sir) Lec-12,13 12 Short notes (Balki sir) Lec-12,13 13	Short notes (Balki sir) Lec-12,13 14 Short notes (Balki sir) Lec-12,13 (contd.) 15 Part II — Short Notes (Q.M), Pages 1 to 10 17 Short notes (Q.M) — Page 1 17 Short notes (Q.M) — Page 2 18 Short notes (Q.M) — Page 3 19 Short notes (Q.M) — Page 4 20 Short notes (Q.M) — Page 5 21 Short notes (Q.M) — Page 6 22 Short notes (Q.M) — Page 7 23 Short notes (Q.M) — Page 7 (contd.) 24 Short notes (Q.M) — Page 8 25 Short notes (Q.M) — Page 9 26 Short notes (Q.M) — Page 10 27
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Part I — Short Notes (Q.M) [Balki sir], Lec-6 to Lec-13

(Short Notes) (Q.M) [Balki sir] Lec-6

To find $\langle A \rangle$ where A can be any arbitrary operator.

$$\langle A \rangle = \langle A \rangle (t) = \frac{\langle \Psi(t) | A | \Psi(t) \rangle}{\langle \Psi(t) | \Psi(t) \rangle}$$

First find $\hat{A} |\Psi(t)\rangle$ then take inner product with unchanged original state $\langle \Psi(t) |$.

Poisson Brackets $\{A, B\}$ are replaced by commutator brackets

$$\{A, B\} \longrightarrow \frac{[A, B]}{i(P.q)}$$

Since $\hat{A}, \hat{B}$ are hermitian, $[A, B]$ should also be hermitian

$$(AB) = (AB - BA)^\dagger = BA - AB$$

To make it hermitian we divide by 'i' so that $\frac{[A, B]}{i}$ is hermitian.

where $P.q$ abbreviation for physical quantity is choosen so that $\frac{[A, B]}{i(P.q)}$ is dimensionless.

The Schrödinger equation :-

The equation which sits at heart of Quantum mechanical formalism of microscopic world / our world is given by

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$

(This is one of postulate of QM)

$$|\Psi(t)\rangle = e^{\frac{-i\hat{H}(t-t_0)}{\hbar}} |\Psi(t_0)\rangle$$

Say, $t_0 = 0$. $|\Psi(t)\rangle = e^{\frac{-i\hat{H}t}{\hbar}} |\Psi(0)\rangle$

where we can say that the time evolution operator $e^{\frac{-i\hat{H}t}{\hbar}}$ takes state $|\Psi(0)\rangle$ to state $|\Psi(t)\rangle$.

$$\langle \Psi(t) | \Psi(t) \rangle = \left\langle \Psi(0) \left| e^{\frac{i\hat{H}t}{\hbar}} e^{\frac{-i\hat{H}t}{\hbar}} \right| \Psi(0) \right\rangle$$

$$\langle \Psi(t) | \Psi(t) \rangle = \langle \Psi(0) | \Psi(0) \rangle$$

← length of vector is preserved

Operator $e^{-i\hat{H}t/\hbar}$ has many names.

- 1) Time development operator
- 2) Evolution operator.
- 3) Propagator

It is unitary operator.
 $UU^\dagger = U^\dagger U = I$
 i.e. $U^\dagger = U^{-1}$

$$U(t_2, t_0) = U(t_2, t_1) \cdot U(t_1, t_0)$$

It follows semi-group property.

Operator 'A' can be diagonalized with similarity transformation if $[A, A^\dagger] = 0$

If $\hat{A}$ is Q.M. operator $A^\dagger = A$ (hermitian) $(A, A^\dagger) = 0$ (always)

Operators $\hat{A}$ & $\hat{B}$ share common set of eigen values and eigen vectors if $[A, B] = 0$

→ If $[\hat{A}, \hat{B}] \neq 0 \Rightarrow$ A common set of eigen states does not exist. But they may share 1 or 2 common eigen states/values.

→ If $[A, B] = 0 \Rightarrow \exists$ a common set of eigen states. ex $[P^2, I] = 0$

$$\hat{P} = \text{projection operator} = |\phi\rangle \langle\phi|$$

$$P^2 = |\phi\rangle \langle\phi| \phi\rangle \langle\phi| = P$$

$$[P^2, I] = [P, I] = 0$$

Every function is eigen state of 'I', but not of $\hat{P}$. Eigen set of $I = \{\text{odd } f^n, \text{ even } f^n, \text{ others}\}$

Eigen set of $P = \{\text{odd fun, even fun}\}$

(Short Notes) (Q.M) [Balki sir] Lec-7,8

Continuous basis :

We can also have continuous basis set to express state vector of hilbert space.

$\hat{x}$ & $\hat{p}$ can have infinite eigen values and thus infinite continuous eigen states for $\hat{x}$ and $\hat{p}$.

$$\hat{x} |x'\rangle = x' |x'\rangle \quad \hat{p} |p'\rangle = p' |p'\rangle$$

Completeness

$$\left(\int_{-\infty}^{\infty} |x\rangle \langle x| dx \right) |\Psi\rangle = |\Psi\rangle$$

$$\left(\int_{-\infty}^{\infty} |p\rangle \langle p| dp \right) |\Psi\rangle = |\Psi\rangle$$

Eigen states of $\hat{x}$ can be orthonormalized.

$$\left. \begin{aligned} \langle x|x'\rangle &= \delta(x-x') \\ \langle p|p'\rangle &= \delta(p-p') \end{aligned} \right\} \text{orthonormality}$$

We can expand $|\Psi\rangle$ in $\{|x\rangle\}$ basis as well as in $\{|p\rangle\}$ basis set.

$$|\Psi\rangle = \int dx |x\rangle \underbrace{\langle x|\Psi(t)\rangle}_{\psi(x,t)}$$

$$|\Psi\rangle = \int_{-\infty}^{\infty} dp |p\rangle \underbrace{\langle p|\Psi(t)\rangle}_{\tilde{\psi}(p,t)}$$

State of Quantum mechanical system is some Abstract vector which lives in seperable Hilbert space.

$|\Psi\rangle$ can take any look depending upon the chosen basis set. We can also express $|\Psi(t)\rangle$ in basis set formed by eigen states of Total Energy operator ($\hat{H}$)

$$\hat{H} |\phi_n\rangle = E_n |\phi_n\rangle$$

$$|\Psi(t)\rangle = \sum_{n=0}^{\infty} c_n |\phi_n(t)\rangle; \quad c_n = \langle \phi_n(t) | \Psi \rangle$$

where, $|\phi_n(t)\rangle = e^{\frac{-i\hat{H}t}{\hbar}} |\phi_n(0)\rangle = e^{\frac{-iE_n t}{\hbar}} |\phi_n(0)\rangle$

let us try to express $\psi(x, t)$ in form of $\tilde{\psi}(p)$.

$$\psi(x, t) = \langle x | \Psi(t) \rangle = \langle x | \int dp |p\rangle \langle p | \Psi \rangle$$

$$\psi(x, t) = \int_{-\infty}^{\infty} dp \langle x | p \rangle \tilde{\psi}(p)$$

Similarly

$$\tilde{\psi}(p, t) = \langle p | \Psi(t) \rangle = \langle p | \int_{-\infty}^{\infty} dx |x\rangle \langle x | \Psi \rangle$$

$$\tilde{\psi}(p, t) = \int_{-\infty}^{\infty} \langle p | x \rangle \bar{\psi}(x, t) dx$$

$$\langle x | \hat{p} | \Psi \rangle = -i\hbar \frac{\partial}{\partial x} \langle x | \Psi \rangle$$

$$\langle x | \hat{p} | p \rangle = p \langle x | p \rangle = -i\hbar \frac{\partial}{\partial x} \langle x | p \rangle$$

$$\langle x | p \rangle \propto e^{+ipx/\hbar} \quad \{ \langle x | [x, p] | x' \rangle = i\hbar \langle x | x' \rangle \}$$

$$\langle p|\hat{x}|\Psi\rangle = +i\hbar \frac{\partial}{\partial p} \langle p|\Psi\rangle$$

for (3D)

$$\langle r|\hat{p}|\Psi\rangle = -i\hbar \nabla \psi(r, t)$$

$$\langle p|\hat{r}|\Psi\rangle = +i\hbar \nabla \tilde{\psi}(p, t)$$

$$\langle r|p\rangle \propto e^{i\vec{p}\cdot\vec{r}/\hbar} \quad \langle p|r\rangle \propto e^{-i\vec{p}\cdot\vec{r}/\hbar}$$

Schrödinger eqⁿ in $\{|r\rangle\}$ basis.

$$\langle r|i\hbar \frac{d}{dt} |\Psi(t)\rangle = \langle r|\hat{H}|\Psi(t)\rangle$$

$$i\hbar \frac{\partial}{\partial t} \psi(r, t) = \left\langle r \left| \frac{p^2}{2m} + \hat{V}(r) \right| \Psi(t) \right\rangle$$

$$i\hbar \frac{\partial}{\partial t} \psi(r, t) = \frac{-\hbar^2}{2m} \nabla^2 \psi(r, t) + V(r) \psi(r, t)$$

↘ Not true if $V(r)$ is Complex.

$$\begin{cases} A|\lambda\rangle = \lambda|\lambda\rangle \\ \langle\lambda|A^\dagger = \langle\lambda|\lambda^* \\ \text{If } (A^\dagger = A) \quad \langle\lambda|\hat{A} = \lambda^* \langle\lambda| \end{cases}$$

If $V(r^\dagger) = V(r)$ i.e. e.v. of V are real

(Short Notes) (Q.M) [Balki sir] Lec-8, 9

Continued :

If eigen values of $V(r^\dagger)$ is imaginary then $\hat{H}$ won't be hermitian and probabilities won't be conserved with time.

$$\langle \Psi(t)|\Psi(t)\rangle \neq \langle \Psi(0)|\Psi(0)\rangle$$

Such kind of complex potentials are used where if some particles are targeted to some target and target absorbs some of them.

If $V(r^\dagger) = V(r)$ (real potential function)

$$i\hbar \frac{\partial}{\partial t} \psi(r, t) = \frac{-\hbar^2}{2m} \nabla^2 \psi(r, t) + V(r) \psi(r, t)$$

Stationary states / Eigen states/vector of $\hat{H} :-$

Then, $H|\phi_n\rangle = E_n|\phi_n\rangle$ where $|\phi_n\rangle$ is eigen vector of $\hat{H}$ with eigen value E_n .

$$\Rightarrow i\hbar \frac{d}{dt} |\phi_n\rangle = \hat{H} |\phi_n\rangle = E_n |\phi_n\rangle$$

$$|\phi_n(t)\rangle = e^{-iE_n t/\hbar} |\phi_n(0)\rangle$$

we can see that these states do not change with time, so called stationary states.

so a general state of system $|\Psi(t)\rangle$ can be expanded a linear combination of stationary state.

The solⁿ of Schrödinger eqⁿ

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = H |\Psi(t)\rangle$$

is

$$|\Psi(t)\rangle = \sum_{n=0}^{\infty} c_n |\phi_n(t)\rangle = \sum_{n=0}^{\infty} \underbrace{c_n e^{\frac{-iE_n t}{\hbar}}}_{c_n(t)} |\phi_n(0)\rangle$$

$$|\Psi(t)\rangle = \sum_{n=0}^{\infty} c_n(t) |\phi_n(0)\rangle \quad - \textcircled{1}$$

So our basis set $\{|\phi_n(0)\rangle\}$ is time independent where as $c_n(t) = \langle \phi_n(0)|\Psi(t)\rangle$ is time dependent.

All Quantum Interference phenomena occur due to superposition of different energy states.

We can write stationary states in $\{|r\rangle\}$ basis

$$\langle r|\phi_n(t)\rangle = \langle r|e^{\frac{-iE_n t}{\hbar}} \phi_n(t)\rangle$$

$$\phi_n(r, t) = e^{\frac{-iE_n t}{\hbar}} \phi_n(r, 0)$$

Then eq $\textcircled{1}$ becomes in $\{|r\rangle\}$ basis.

$$\langle r|\Psi(t)\rangle = \sum_{n=0}^{\infty} c_n(t) \langle r|\phi_n(0)\rangle$$

$$\Psi(r, t) = \sum_{n=0}^{\infty} c_n(t) \phi_n(r, 0)$$

our Schrödinger eqⁿ becomes,

$$i\hbar \frac{\partial}{\partial t} \Psi(r, t) = \frac{-\hbar^2}{2m} \nabla^2 \Psi(r, t) + V(r) \Psi(r, t)$$

for stationary states,

$$i\hbar \frac{\partial}{\partial t} \phi_n(r, t) = \frac{-\hbar^2}{2m} \nabla^2 \phi_n(r, t) + V(r) \phi_n(r, t)$$

using, $\{\phi_n(r, t) = e^{-iE_n t/\hbar} \phi_n(r, 0)\}$

we have

$$i\hbar \phi_n(r, 0) e^{-\frac{iE_n t}{\hbar}} \left(\frac{-iE_n}{\hbar} \right) = H e^{-\frac{iE_n t}{\hbar}} \phi_n(r, 0)$$

$$E_n \phi_n(r, 0) = H \phi_n(r, 0)$$

or

$$\frac{-\hbar^2}{2m} \nabla^2 \phi_n(r, 0) + V(r) \phi_n(r, 0) = E_n \phi_n(r, 0)$$

lets just write $\phi_n(r, 0) = \phi_n(r)$

$$\boxed{\nabla^2 \phi_n(r) + \frac{2m(E_n - V(r))}{\hbar^2} \phi_n(r) = 0}$$

This equation is used to find time inde-pendent basis set $\{\langle r | \phi_n(0) \rangle\}$ or $\{\phi_n(r, 0)\}$ and often called as Time Independent Schrödinger eqⁿ. (as it does not have any time dependent term).

(Short Notes) (Q.M) [Balki sir] Lec-9,10

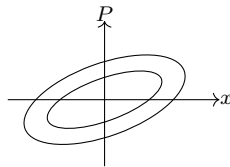
Bohr quantization for periodic orbits :-

we know that

$$\Delta x \Delta p \geq \hbar/2$$

$$dA = dx dp \sim h$$

$$\text{So } \oint dA = nh \Rightarrow \oint P dx = nh$$



↗ Total area is quantized of uncertainty principle.

Finding $\langle A \rangle$ in given stationary state $|\phi\rangle$

$$\langle A \rangle_n = \frac{\langle \phi_n | \hat{A} | \phi_n \rangle}{\langle \phi_n | \phi_n \rangle} \quad \text{If } \langle \phi_n | \phi_n \rangle = 1$$

then, $\langle A \rangle = \langle \phi_n | A | \phi_n \rangle$

we know,

$$|\phi\rangle = I |\phi_n\rangle = \int |x'\rangle \langle x' | \phi_n \rangle dx'$$

$$\langle \phi_n | I = \int dx \langle \phi_n | x \rangle \langle x |$$

$$\text{So } \langle A \rangle_n = \langle \phi_n | \hat{A} | \phi_n \rangle = \int dx \phi_n^*(x) \langle x | \hat{A} \int |x'\rangle \phi_n(x') dx'$$

$$\left\{ \langle \hat{A} \rangle_n = \int dx \phi_n^*(x) \int \langle x | \hat{A} | x' \rangle \phi_n(x') dx' \right\}$$

example : If $\hat{A} = \hat{x}$

$$\begin{aligned}\langle \hat{x} \rangle_n &= \int dx \phi_n^*(x) \int \langle x | \hat{x} | x' \rangle \phi_n(x') dx' \\ &= \int dx \phi_n^*(x) \int x' \langle x | x' \rangle \phi_n(x') dx' \\ &= \int dx \phi_n^*(x) \underbrace{\left(\int x' \delta(x' - x) \phi_n(x') dx' \right)}_{=I}\end{aligned}$$

$$\begin{cases} \langle x | x' \rangle = \langle x' | x \rangle^* = \langle x' | x \rangle \quad (\text{real}) \\ \delta(x - x') = \delta(x' - x) \end{cases}$$

$$\langle x \rangle_n = \int_{-\infty}^{\infty} dx \phi_n^*(x) x \phi_n(x)$$

$$\text{as } I = \begin{cases} 0 & x \neq x' \\ x \phi_n(x) & x = x' \end{cases}$$

Similarly

$$\langle \hat{p} \rangle_n = \int dx \phi_n^*(x) \left(-i\hbar \frac{\partial}{\partial x} \langle x | x' \rangle \right) \phi_n(x') dx'$$

$$\left\{ \langle p \rangle_n = \int \phi_n^*(x) \left(-i\hbar \frac{\partial}{\partial x} \right) \phi_n(x) dx \right\}$$

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

↓ std. deviation / uncertainty in $\hat{x}$.

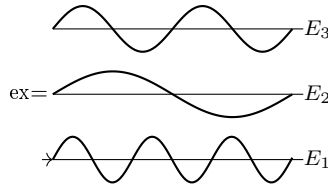
$$\stackrel{\text{or}}{=} \Delta x = \sqrt{\langle (x - \langle x \rangle)^2 \rangle}$$

Similarly

$$\delta P = \sqrt{\langle p^2 \rangle - \langle p \rangle^2}$$

“Symmetry implies degeneracy”

More curvature $\Rightarrow$ more energy



$$(E_1 > E_3 > E_2)$$

$$\text{This can be seen from } \underbrace{\frac{-\hbar^2}{2m} \frac{\partial^2 \phi}{\partial x^2}}_{(\text{curvature})} = \underbrace{E \phi}_{(\text{energy})}$$

We know that for a particle of mass m in potential $V(r)$ [ONLY for $V(r) \neq 0$]

$$\hat{H} = \frac{p^2}{2m} + V(r)$$

$$[H, r] \neq 0 \quad \text{and} \quad \left[H, \frac{p^2}{2m} \right] \neq 0 \quad - \textcircled{1}$$

$$[H, V(r)] \neq 0 \quad - \textcircled{2} \quad \Big| \quad [V(r), r] = 0 \quad - \textcircled{3}$$

Eqⁿ (1) implies that particle does not have simultaneous Total energy & K.E at a given distance ‘ r .’

Eqⁿ (2) $\Rightarrow$ Notion of potential energy for given total energy is no more.

Eqⁿ (3) $\Rightarrow$ we can talk for potential energy at given position r .

Short notes (Balki sir) Lec-11

Nodes vs Energy :- (1-D Infinite well)

we know,

$$\phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad \text{for } 0 \leq x \leq L$$

we can find ‘ E_n ’ from eigen value equation for ‘ H ’.

$$H |\phi_n\rangle = E_n |\phi_n\rangle$$

$$\langle x | H | \phi_n \rangle = E_n \langle x | \phi_n \rangle$$

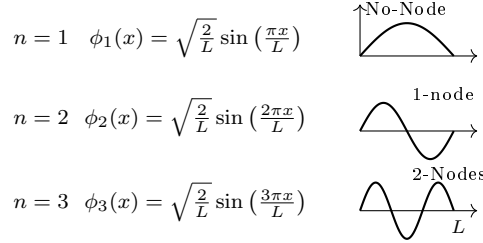
$$\frac{-\hbar^2}{2m} \frac{d^2}{dx^2} \phi_n(x) = E_n \phi_n(x) \quad - \textcircled{1}$$

$$\left\{ \begin{array}{l} \phi'(x) = \sqrt{\frac{2}{L}} \frac{n\pi}{L} \cos\left(\frac{n\pi x}{L}\right) \\ \phi''(x) = -\sqrt{\frac{2}{L}} \left(\frac{n\pi}{L}\right)^2 \sin\left(\frac{n\pi x}{L}\right) \end{array} \right\}$$

Using in eqⁿ ① we have

$$\frac{\hbar^2}{2m} \left(\frac{n^2 \pi^2}{L^2} \right) (\phi_n(x)) = E_n (\phi_n(x))$$

$$\boxed{E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}} \quad (E_n \propto n^2)$$



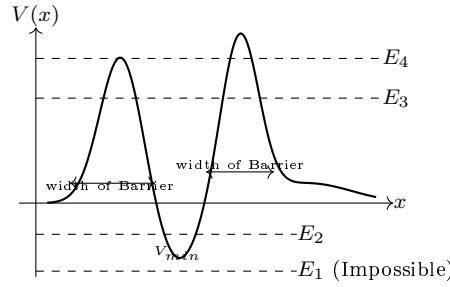
no. of nodes (m) = $n - 1$

or ($n = m + 1$)

$$\Rightarrow E_n = E_{m+1} = \frac{(m+1)^2 \pi^2 \hbar^2}{2mL^2}$$

as number of nodes increases, $\phi''(x)$ contributes to $E_n \left\{ \frac{-\hbar^2}{2m} \phi''(x) = E_n \phi(x) \right\}$ and that is why energy increases.

Study of general potential :



E_1, E_2, E_3, E_4 are arbitrary chosen possibilities of total energy of system.

$$\textcircled{1} E_1 < V_{min} \Rightarrow \langle H \rangle < \langle V \rangle \Rightarrow \langle H \rangle - \langle V \rangle < 0$$

$$\text{But } \langle H \rangle = \left\langle \frac{p^2}{2m} \right\rangle + \langle V \rangle$$

$$\Rightarrow \left\langle \frac{p^2}{2m} \right\rangle = \langle H \rangle - \langle V \rangle = -ve$$

But

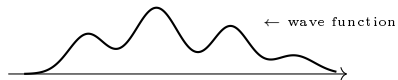
$$\left\langle \frac{p^2}{2m} \right\rangle = \frac{\langle \psi | P P | \psi \rangle}{2m} = \frac{\langle \psi | P^\dagger P | \psi \rangle}{2m} = \frac{\langle P^\dagger \psi | P \psi \rangle}{2m}$$

$\uparrow$
Norm > 0

So ($E_1 < V_{min}$) is not possible.

$$\textcircled{2} V_{min} < E < 0 \text{ (} E_2 \text{) :-}$$

Most of the time particle will be found inside well, but $\exists$ finite non-zero probability that it will tunnel and escape.

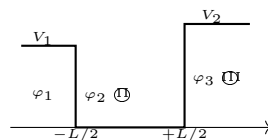


But $\exists$ bound states as well depending upon E_2 .

$$\textcircled{3} (E > 0) \text{ (} E_3 \text{ or } E_4 \text{)}$$

since, $E_3(\infty)$ or $E_4(\infty) > V(\infty)$, so particle will always escape to infinity and bound state does not exist for such energy values.

Bound states in Finite square well



For $E > V_1$ & $E > V_2$ we only have scattering states. To have Bound states we need $E < V_1$ & $E < V_2$

① Region (I) :-

$$\frac{d^2 \varphi_1}{dx^2} - k^2 \varphi_1 = 0 \quad k^2 = \frac{2m(V_1 - E)}{\hbar^2}$$

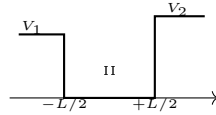
$$\varphi_1 = Ae^{-kx} + Be^{kx}$$

as $(x \rightarrow -\infty) \varphi_1 \rightarrow 0 \Rightarrow (A = 0)$

$$\underline{\varphi_1 = Be^{kx}} \quad (x < -L/2)$$

Short notes (Balki sir) Lec-11

② Region II :-



$$\frac{d^2 \varphi_2}{dx^2} + \frac{2m(E - 0)}{\hbar^2} \varphi_2 = 0$$

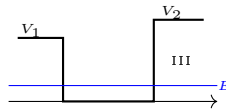
$$(\partial^2 + l^2) \varphi_2 = 0 \quad (l^2 = 2mE/\hbar^2)$$

$$(\varphi_2 = ae^{ilx} + be^{-ilx})$$

or $\varphi_2 = a \sin(lx) + b \cos lx$

or $\{\varphi_2 = a \sin(lx + \theta)\}$

Region III :



$$\frac{d^2 \varphi_3}{dx^2} + \frac{2m(E - V_2)}{\hbar^2} \varphi_3 = 0$$

Since, $E < V_2 \Rightarrow (E - V_2 < 0)$

$$(\partial^2 - m^2) \varphi_3 = 0 \quad \left\{ m^2 = \frac{2m(V_2 - E)}{\hbar^2} \right\}$$

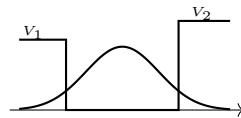
$$\varphi_3(x) = ce^{mx} + de^{-mx}$$

need $\varphi_3 \rightarrow 0$ as $x \rightarrow +\infty \Rightarrow c = 0$

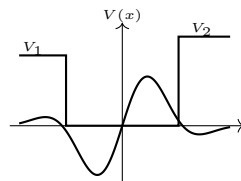
$$\boxed{\varphi_3(x) = de^{-mx}}$$

Since $V(x)$ is finite, we can't have $\varphi''(x)$ as infinite, so we need continuity in $\varphi'(x)$ as well as in $\varphi(x)$

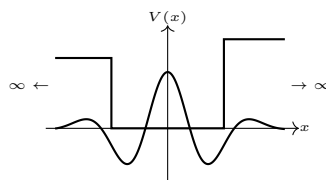
Ground state : (0-nodes)



1st excited state : (1-node)



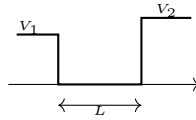
2nd excited state : (2-nodes)



No. of Bound states :

(1) For $E > V(\infty)$: No Bound states exist

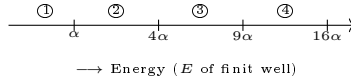
(2) For $E < V(\infty)$



we know that,

$$\frac{2mE_n}{\hbar^2} = k^2 = \frac{n^2\pi^2}{L^2} \quad \text{for (1-D) } \infty\text{-well}$$

$$\left(E_n = \frac{n^2\hbar^2\pi^2}{2mL^2}\right) = n^2\alpha \quad \left(\alpha = \frac{\pi^2\hbar^2}{2mL^2}\right)$$



(1) $E < \alpha$

(2) $\alpha < E < 4\alpha$

(3) $4\alpha < E < 9\alpha$

(4) $9\alpha < E < 16\alpha$

$\vdots$

No of Bound states

$\frac{1}{2}$

2

3

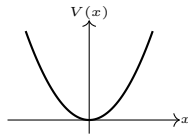
4

Eigen functions of symmetric potential will be either even or odd functions

Linear Harmonic Oscillator :-

$$H = \frac{P^2}{2m} + \frac{1}{2}m\omega^2x^2$$

$$(V(x) = \frac{1}{2}m\omega^2x^2)$$



$$H = \hbar\omega \left(a^\dagger a + \frac{1}{2}\right)$$

$$a = \frac{x}{\sqrt{2\hbar/m\omega}} + \frac{ip}{\sqrt{2\hbar m\omega}} = \frac{x}{x_0} + \frac{ip}{p_0}$$

$$a^\dagger = \frac{x}{\sqrt{2\hbar/m\omega}} - \frac{ip}{\sqrt{2\hbar m\omega}} = \frac{x}{x_0} - \frac{ip}{p_0}$$

where $[a, a^\dagger] = 1$

then, $E_n = \hbar\omega \left(n + \frac{1}{2}\right)$ ($n = 0, 1, 2, 3, \dots$)

and,

$$\phi_n(x) = A_n e^{-\left(\frac{x}{x_0}\right)^2} H_n(x/x_0)$$

$$\tilde{\phi}_n(p) = B_n e^{-\left(\frac{p}{p_0}\right)^2} H_n(p/p_0)$$

Short notes (Balki sir) Lec-11,12

we saw that

$$\phi_n(x) = A_n e^{-(x/x_0)^2} H_n(x/x_0)$$

are stationary states for S.H.O.

where $H_n(x)$ are Hermite polynomials.

$$H_n(x) = e^{x^2} \left(\frac{-d}{dx}\right)^n e^{-x^2}$$

$$H_0(x) = e^{x^2} e^{-x^2} = 1$$

$$H_1(x) = e^{x^2} (+2x) e^{-x^2} = 2x$$

$$H_2(x) = e^{x^2} (-2e^{-x^2} + 4x^2 e^{-x^2})$$

$$H_2(x) = -2 + 4x^2$$

a — lowering / annihilation operator

$a^\dagger$ — Raising / creation operator

$$a|n\rangle = c_n|n-1\rangle$$

$$\langle n|a^\dagger = \langle n-1|c_n^*$$

$$\langle n|a^\dagger a|n\rangle = |c_n|^2$$

$$\langle n|\hat{N}|n\rangle = |c_n|^2$$

$$c_n = \sqrt{n}$$

So,

$$a|n\rangle = \sqrt{n}|n-1\rangle$$

$$a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

Some useful results

$$1) \sum_{n=1}^n n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$2) \sum_{n=1}^n n^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$3) \sum_{n=1}^n n^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

$$= (\sum n)^2$$

$$4) \sum_{n=1}^n (2n)^3 = 2^3 \sum_{n=1}^n n^3 = 8 \left(\frac{n(n+1)}{2}\right)^2 = 8 (\sum n)^2$$

$$5) (x+y)^4 = \left(\frac{4!}{4!}\right)x^4 + \left(\frac{4!}{4!}\right)y^4 + \left(\frac{4!}{2!2!}\right)x^2y^2$$

$$+ \left(\frac{4!}{1!3!}\right)xy^3 + \left(\frac{4!}{3!1!}\right)x^3y$$

Ground State of SHO :

$$a|0\rangle = 0$$

$$\Rightarrow \langle x|a|0\rangle = 0$$

$$\left\langle x \left| \frac{\hat{x}}{x_0} + \frac{i\hat{p}}{\hbar} \right| 0 \right\rangle = 0$$

$$\frac{1}{x_0} \langle x|\hat{x}|0\rangle + \frac{i}{p_0} \langle x|\hat{p}|0\rangle = 0$$

$$\frac{x}{x_0} \phi_0(x) + \frac{i}{p_0} \left(-i\hbar \frac{\partial}{\partial x} \phi_0(x) \right) = 0$$

Solve to get $\phi(x)$:

$$\left\{ \frac{x}{x_0} \phi_0(x) + \frac{\hbar}{p_0} \frac{\partial \phi_0(x)}{\partial x} = 0 \right\}$$

$$\begin{cases} \phi_0(x) = \langle x|0\rangle \\ \phi_1(x) = \langle x|1\rangle \\ \phi_2(x) = \langle x|2\rangle \\ \phi_n(x) = \langle x|n\rangle \end{cases}$$

To find $\phi_1(x) = \langle x|1\rangle$

$$a|n\rangle = \sqrt{n}|n-1\rangle$$

$$a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

So, $a^\dagger|0\rangle = 1|1\rangle$

$$\langle x|a^\dagger|0\rangle = \langle x|1\rangle = \phi_1(x)$$

$$\left\langle x \left| \frac{x}{x_0} + \frac{ip}{p_0} \right| 0 \right\rangle = \phi_1(x)$$

$$\left\{ \frac{x}{x_0} \phi_0(x) + \frac{i}{p_0} \left(-i\hbar \frac{\partial}{\partial p} \phi_0(x) \right) = \phi_1(x) \right\}$$

$\phi_1(x)$ will turn out to be normalized.

Some results :

$$1) \langle x \rangle_n = \langle n|\hat{x}|n\rangle = \left\langle n \left| \frac{x_0(a+a^\dagger)}{2} \right| n \right\rangle = 0$$

$$2) \langle p \rangle_n = \langle n | \hat{p} | n \rangle = \left\langle n \left| \frac{p_0}{2i} (a - a^\dagger) \right| n \right\rangle = 0$$

$$\begin{aligned} 3) \langle x^2 \rangle_n &= \langle n | \hat{x}^2 | n \rangle = \left\langle n \left| \frac{x_0^2}{4} (a + a^\dagger)^2 \right| n \right\rangle \\ &= \left\langle n \left| \frac{x_0^2}{4} (aa^\dagger + a^\dagger a) \right| n \right\rangle \\ &= \frac{x_0^2}{4} \langle n | N + 1 + N | n \rangle \end{aligned}$$

$$\langle x^2 \rangle_n = \left(n + \frac{1}{2} \right) \frac{x_0^2}{2} = \frac{1}{2} \left(\frac{2\hbar}{2m\omega} \right) \left(n + \frac{1}{2} \right)$$

$$\begin{aligned} 4) \langle p^2 \rangle_n &= \langle n | \hat{p}^2 | n \rangle = \left\langle n \left| \frac{p_0^2}{4i^2} (a - a^\dagger)^2 \right| n \right\rangle \\ &= \frac{-p_0^2}{4} \left\langle n \left| -aa^\dagger - a^\dagger a \right| n \right\rangle \\ &= \frac{p_0^2}{4} \langle n | 2N + 1 | n \rangle = \hbar m \omega \left(n + \frac{1}{2} \right) \end{aligned}$$

Short notes (Balki sir) Lec-12,13

Eigen-states of $(a^\dagger)$:

$$a^\dagger |\lambda\rangle = \lambda |\lambda\rangle$$

where $|\lambda\rangle$ can be expanded in $\{|n\rangle\}$ basis :- $|\lambda\rangle = \sum_{n=0}^{\infty} c_n |n\rangle$

$$a^\dagger |\lambda\rangle = \sum_{n=0}^{\infty} c_n a^\dagger |n\rangle = \sum_{n=0}^{\infty} c_n \sqrt{n+1} |n+1\rangle \quad - \textcircled{1}$$

$$\text{also, } a^\dagger |\lambda\rangle = \lambda |\lambda\rangle = \sum_{n=0}^{\infty} \lambda c_n |n\rangle \quad - \textcircled{2}$$

$$\Rightarrow \sum_{n=0}^{\infty} c_n \lambda |n\rangle = \sum_{n=0}^{\infty} c_n \sqrt{n+1} |n+1\rangle$$

which gives,

$$\begin{aligned} &\lambda (c_0 |0\rangle + c_1 |1\rangle + c_2 |2\rangle + \dots) \\ &= c_0 |1\rangle + c_1 \sqrt{2} |2\rangle + c_2 \sqrt{3} |3\rangle + \dots \\ &\Rightarrow \lambda c_0 = 0, \quad \lambda c_1 = c_0, \quad \lambda c_2 = c_1 \sqrt{2}, \\ &\dots \Rightarrow c_0 = 0 = c_1 = c_2 = c_3 \dots \\ &\Rightarrow |\lambda\rangle \text{ d.n.e.} \end{aligned}$$

Eigen states of a :

$$\hat{a} |\alpha\rangle = \alpha |\alpha\rangle$$

where, $|\alpha\rangle = \sum_{n=0}^{\infty} c_n |n\rangle$

$$\Rightarrow a |\alpha\rangle = \sum_{n=0}^{\infty} c_n a |n\rangle = \sum_{n=1}^{\infty} c_n \sqrt{n} |n-1\rangle$$

$$\text{also, } a |\alpha\rangle = \alpha |\alpha\rangle = \sum_{n=0}^{\infty} \alpha c_n |n\rangle$$

$$\Rightarrow \sum_{n=1}^{\infty} c_n \sqrt{n} |n-1\rangle = \sum_{n=0}^{\infty} \alpha c_n |n\rangle$$

$$\Rightarrow (c_1 |0\rangle + c_2 \sqrt{2} |1\rangle + c_3 \sqrt{3} |2\rangle + \dots)$$

$$= \alpha (c_0 |0\rangle + c_1 |1\rangle + c_2 |2\rangle + \dots)$$

$$\begin{aligned} &c_1 = \alpha c_0 \\ &c_2 \sqrt{2} = \alpha c_1 \quad c_2 = \frac{1}{\sqrt{2}} \alpha^2 c_0 \\ \Rightarrow &c_3 \sqrt{3} = \alpha c_2 \quad c_3 = \frac{1}{\sqrt{3 \cdot 2}} \alpha^3 c_0 \\ &\vdots \\ &c_n \sqrt{n} = \alpha c_{n-1} \quad c_n = \frac{1}{\sqrt{n!}} \alpha^n c_0 \end{aligned}$$

$$\Rightarrow |\alpha\rangle = \sum_{n=0}^{\infty} c_n |n\rangle = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n!}} \alpha^n c_0 |n\rangle$$

$$\left\{ |\alpha\rangle = c_0 \sum_{n=0}^{\infty} \frac{\alpha^n |n\rangle}{\sqrt{n!}} \right\}$$

$$\langle \alpha | \alpha \rangle = 1 \Rightarrow |c_0|^2 \sum_{n=0}^{\infty} \frac{(|\alpha|^2)^n}{n!} = 1$$

$$|c_0|^2 e^{|\alpha|^2} = 1 \Rightarrow \left(|c_0| = e^{-\frac{|\alpha|^2}{2}} \right)$$

So

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

$$a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$|1\rangle = a^\dagger |0\rangle \quad |2\rangle = \frac{1}{\sqrt{2}} a^\dagger |1\rangle = \frac{(a^\dagger)^2}{\sqrt{2}} |0\rangle$$

$$|3\rangle = \frac{1}{\sqrt{3}} a^\dagger |2\rangle = \frac{(a^\dagger)^3}{\sqrt{3!}} |0\rangle \quad \vdots$$

$$\left(|n\rangle = \frac{1}{\sqrt{n!}} (a^\dagger)^n |0\rangle \right)$$

using, we get,

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n (a^\dagger)^n}{n!} |0\rangle$$

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \left(\sum_{n=0}^{\infty} \frac{(\alpha a^\dagger)^n}{n!} \right) |0\rangle$$

So

$$\left\{ |\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} e^{\alpha a^\dagger} |0\rangle \right\}$$

↖
operator

Normalized eigen states of lowering operator.

Note : All numbers of complex plane (α) are eigen values of annihilation operator.

States $|\alpha\rangle$ are not orthonormal and forms over-complete space of states.

$$\left(\int d\alpha |\alpha\rangle \langle \alpha| = \pi I \right)$$

$$\begin{aligned} \langle x | \alpha \rangle &= \langle x | \sum_{n=0}^{\infty} c_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle \\ &= \langle x | e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle \\ &= e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \phi_n(x) \end{aligned}$$

$$\alpha(x) = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \left(A_n e^{-\frac{x^2}{2}} H_n(x/x_0) \right)$$

Matrix representations :

$$a_{\ell m} = \langle \ell | a | m \rangle, \quad a_{\ell m}^\dagger = \langle \ell | a^\dagger | m \rangle$$

$$x_{\ell m} = \langle \ell | x | m \rangle, \quad P_{\ell m} = \langle \ell | P | m \rangle$$

Short notes (Balki sir) Lec-12,13

$$A = \vec{J} \cdot \hat{n}$$

$$\hat{n} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$$

$$\hat{A} = \sin \theta \cos \varphi \hat{J}_x + \sin \theta \sin \varphi \hat{J}_y + \cos \theta \hat{J}_z$$

$$\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$$

$$\begin{aligned} \langle A^2 \rangle = & \left\langle \sin^2 \theta \cos^2 \varphi J_x^2 + \sin^2 \theta \sin^2 \varphi J_y^2 + \cos^2 \theta J_z^2 \right. \\ & + \frac{\sin^2 \theta \sin 2\varphi}{2} J_x J_y + \frac{\sin 2\theta \cos \varphi}{2} J_x J_z \\ & + \frac{\sin 2\varphi}{2} J_y J_x + \frac{\sin 2\theta}{2} \sin \varphi J_y J_z \\ & \left. + \frac{\sin 2\theta}{2} \cos \varphi J_z J_x + \frac{\sin 2\theta}{2} \sin \varphi J_z J_y \right\rangle \end{aligned}$$

(cross terms $\rightarrow 0$)

$$\langle J_x J_y + J_y J_x \rangle = ? \longrightarrow \langle 2J_x J_y - i\hbar J_z \rangle = 2\langle J_x J_y \rangle - i\hbar m\hbar$$

We know, $(J_x J_y - J_y J_x = i\hbar J_z)$

$$= 2\left(\frac{i\hbar}{2} m\hbar\right) - i\hbar m\hbar = 0$$

$$\langle J_x^2 \rangle = \langle J_y^2 \rangle = \frac{\langle J^2 \rangle - \langle J_z^2 \rangle}{2} = \frac{j(j+1)\hbar^2 - m^2\hbar^2}{2}$$

$$\langle A^2 \rangle = \sin^2 \theta \langle J_x^2 \rangle + \cos^2 \theta \langle J_z^2 \rangle = \sin^2 \theta \langle J_x^2 \rangle + \cos^2 \theta m^2 \hbar^2$$

$$\begin{aligned} \langle J_x J_y \rangle &= \left\langle \left(\frac{J_+ + J_-}{2} \right) \left(\frac{J_+ - J_-}{2i} \right) \right\rangle \\ &= \left\langle \frac{J_+^2 - J_+ J_- + J_- J_+ - J_-^2}{4i} \right\rangle \\ &= \frac{\langle [J_-, J_+] \rangle}{4i} = \frac{\langle [(J_x - iJ_y), (J_x + iJ_y)] \rangle}{4i} \\ &= \left(\frac{\cancel{0} + i^2 \hbar J_z + i i J_z \hbar}{4i} \right) \\ &= \left\langle \frac{-J_z \hbar}{2i} \right\rangle = \left\langle \frac{i\hbar J_z}{2} \right\rangle \\ &= \frac{i\hbar}{2} m\hbar \end{aligned}$$

$$\begin{aligned} \langle J_x J_z \rangle &= \langle jm | J_x J_z | jm \rangle \\ &= m\hbar \langle jm | J_x | jm \rangle = 0 \end{aligned}$$

$$\begin{aligned} \langle J_z J_x \rangle &= \langle jm | J_z J_x | jm \rangle \\ &= m\hbar \langle jm | J_x | jm \rangle = m\hbar(0) = 0 \end{aligned}$$

$$\langle J_z J_y \rangle = \langle J_y J_z \rangle = 0.$$

$$\begin{aligned} \langle A \rangle &= \sin \theta \cos \varphi \cancel{\langle J_x \rangle^0} + \sin \theta \sin \varphi \cancel{\langle J_y \rangle^0} + \cos \theta \langle J_z \rangle \\ &= \cos \theta m\hbar \end{aligned}$$

$$\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$$

$$\Delta A = \sqrt{\sin^2 \theta \langle J_x^2 \rangle} = \sqrt{\langle J_x^2 \rangle} \sin \theta$$

$$\Delta A = \left(\sqrt{\frac{j(j+1) - m^2}{2}} \right) \hbar \sin \theta$$

Short notes (Balki sir) Lec-12,13

This universe is not mine,
I have not created it.

This life itself is not mine,
I am not my creator.

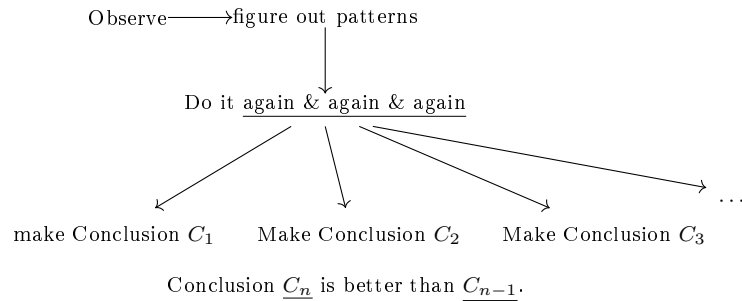
Nobody is creator of anything here.
Every one does is to modify things

for their longevity.

So, Here I am knowing about this universe / Nature or nature of Nature. Training myself from failure to less failure. Making an understanding of World in which I live on.

Just like Artificial machines, I will train my mind to break some unproductive habits, and turn it into an useful machine to understand this nature.

To learn I need to observe, repeat / redo, make conclusions.



Short notes (Balki sir) Lec-12,13

$$\frac{d}{dt} \langle A \rangle = \left\langle \frac{dA}{dt} \right\rangle$$

where $\langle A \rangle = \frac{\langle \psi(0) | A(t) | \psi(0) \rangle}{\langle \psi(0) | \psi(0) \rangle}$

or $\langle A \rangle = \frac{\langle \psi_H | A(t) | \psi_H \rangle}{\langle \psi_H | \psi_H \rangle}$

$$\frac{d}{dt} \langle A \rangle = \frac{\langle \psi(0) | \frac{dA}{dt} | \psi(0) \rangle}{\langle \psi(0) | \psi(0) \rangle}$$

$$\left\{ \frac{d}{dt} \langle A \rangle = \left\langle \frac{dA}{dt} \right\rangle \right\}$$

From classical mech.

For any $A = A(q, p)$

Heisenberg

$$\frac{dA}{dt} = \sum_{i=1}^N \left(\frac{\partial A}{\partial q} \dot{q} + \frac{\partial A}{\partial p} \dot{p} \right) + \frac{\partial A}{\partial t}$$

equation of motion.

$$\frac{dA}{dt} = \{A, H\} + \partial A / \partial t$$

$$\frac{d\hat{A}(t)}{dt} = \frac{[A, H]}{i\hbar} + \partial \hat{A} / \partial t$$

Probability current density associated with probability density :-

$$\nabla \cdot \vec{J} = -\partial \rho / \partial t$$

$$\rho = |\psi|^2, \quad i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle$$

$$\hat{H} = \frac{-\hbar^2}{2m} \frac{d^2}{dx^2} + \hat{V}(x)$$

$$\left\{ \vec{J} = \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) \right\}$$

Evolution of state after collapse :-

suppose we have particle inside a (1-D) box.

$$\psi(x) = \langle x | \Psi \rangle = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

Any style measurement of position will collapse wavefunction to x_0 .

$$\psi(x = x_0) = \delta(x - x_0)$$

Now this state will evolve with time. find $\langle x \rangle(t)$, $\langle x^2 \rangle(t)$, $\langle x^3 \rangle(t) \dots$

lets use Heisenberg picture. So state vector remains fixed $= \delta(x - x_0)$, & $\hat{x} = \hat{x}(t)$

To know, how wavefunction changes let us look how $\langle x \rangle$, $\langle x^2 \rangle(t)$, σ^2 changes with time.

$$-\textcircled{1} \frac{d}{dt} \langle x \rangle = \langle dx/dt \rangle = \left\langle \frac{[x, H]}{i\hbar} \right\rangle$$

$$= \left\langle \frac{\left[x, \frac{p^2}{2m} + V(x) \right]}{i\hbar} \right\rangle$$

$$\frac{d}{dt} \langle x \rangle = \frac{1}{2mi\hbar} \langle 2p i\hbar \rangle = \frac{\langle p \rangle}{m}$$

$$\left\{ \frac{d}{dt} \langle x \rangle = \frac{\langle p \rangle}{m} \right\}$$

$$\begin{aligned} (2) \frac{d}{dt} \langle x^2 \rangle &= \left\langle \frac{dx^2(t)}{dt} \right\rangle = \frac{\langle [x^2, H] \rangle}{i\hbar} \\ &= \left(\frac{1}{i\hbar} \right) \left\langle \left[x^2, \frac{p^2}{2m} + V(x) \right] \right\rangle \\ &= \frac{1}{2mi\hbar} \langle [x^2, p^2] \rangle \\ &= \frac{1}{2mi\hbar} \langle x [x, p^2] + [x, p^2] x \rangle \\ &= \frac{1}{2mi\hbar} \langle x(2pi\hbar) + (2pi\hbar)x \rangle \end{aligned}$$

$$\left\{ \frac{d}{dt} \langle x^2 \rangle = \frac{1}{m} \langle xp + px \rangle \right\}$$

Number
Not an operator

$$\begin{aligned} \frac{d^2}{dt^2} \langle x^2 \rangle &= \frac{1}{m} \left\langle \frac{d}{dt} (xp + px) \right\rangle \\ &= \frac{1}{m} \left\langle \frac{\left[xp + px, \frac{p^2}{2m} + V(x) \right]}{i\hbar} + \frac{\partial}{\partial t} \langle xp + px \rangle \right\rangle \\ &= \frac{1}{mi\hbar} \left\langle \left[xp + px, \frac{p^2}{2m} \right] \right\rangle + \frac{\langle [xp + px, V] \rangle}{mi\hbar} \\ &= \frac{2}{m^2} \langle p^2 \rangle + \frac{\langle [xp + px, V(x)] \rangle}{mi\hbar} \\ &= \frac{2}{m^2} \langle p^2 \rangle + \frac{x[p, V(x)] + [p, V(x)]x}{mi\hbar} \end{aligned}$$

$$\langle \ddot{x^2} \rangle = \frac{2}{m^2} \langle p^2 \rangle + \frac{-2}{m} \frac{dV(x)}{dx}$$

Short notes (Balki sir) Lec-12,13

Heisenberg picture of Q.M.

Observables / operators are time dependent, where as state of quantum mechanical system is fixed.

$$\langle A \rangle(t) = \frac{\langle \psi | A | \psi \rangle}{\langle \psi | \psi \rangle}$$

↓

This time dependence came from time dependent $|\psi(t)\rangle$ in Schrödinger interpretation of Q.M.

$$\langle A \rangle(t) = \frac{\langle \psi_s(t) | \hat{A}_s | \psi_s(t) \rangle}{\langle \psi_s(t) | \psi_s(t) \rangle}$$

$$\left\{ |\psi(t)\rangle = e^{\frac{-iHt}{\hbar}} |\psi(0)\rangle \right\}$$

$$\langle A \rangle(t) = \frac{\langle \psi(0) | e^{\frac{iHt}{\hbar}} A_s e^{\frac{-iHt}{\hbar}} | \psi(0) \rangle}{\langle \psi(0) | \psi(0) \rangle}$$

$$\left\{ \langle A(t) \rangle(t) = \frac{\langle \psi(0) | A(t) | \psi(0) \rangle}{\langle \psi(0) | \psi(0) \rangle} \right\}$$

where,

$$A(t) = A_H(t) = e^{\frac{iH_s t}{\hbar}} A_s e^{-\frac{iH_s t}{\hbar}}$$

$$(t=0) \text{ gives } A_H(0) = A_s$$

So,

$$A(t) = e^{\frac{iH_s t}{\hbar}} A_H(0) e^{-\frac{iH_s t}{\hbar}}$$

$$H(t) = e^{\frac{iH_s t}{\hbar}} H_s e^{-\frac{iH_s t}{\hbar}}$$

also, we know $A_H(0) = A_s$

i.e. $H_H(0) = H_s$

$$\text{So, } \left[e^{\frac{iH_s t}{\hbar}}, H_s \right] = 0$$

$$\Rightarrow H(t) = H_s = H_H(0)$$

$$\Rightarrow \left\{ A_H(t) = e^{\frac{iH t}{\hbar}} A(0) e^{-\frac{iH t}{\hbar}} \right\}$$

$$\begin{aligned} \frac{dA_H(t)}{dt} &= e^{\frac{iH t}{\hbar}} \frac{iH}{\hbar} A(0) e^{-\frac{iH t}{\hbar}} \\ &\quad + e^{\frac{iH t}{\hbar}} A(0) e^{-\frac{iH t}{\hbar}} \left(\frac{-iH}{\hbar} \right) \\ &= \frac{i}{\hbar} H A(t) + A(t) (-iH/\hbar) \end{aligned}$$

$$\left\{ \frac{dA(t)}{dt} = \frac{[A, H]}{i\hbar} \right\}$$

Schrödinger Picture :

Each quantum mechanical system is described by a state vector $|\Psi(t)\rangle$ which evolves in time by

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$

where as, observables are time independent, self adjoint, Hermitian operators.

$$A + f(t) \quad |\psi(t)\rangle = e^{-\frac{iH t}{\hbar}} |\psi(0)\rangle$$

$$\langle A \rangle(t) = \frac{\langle \psi(t) | A | \psi(t) \rangle}{\langle \psi | \psi \rangle} \quad |\psi(t)\rangle = e^{-\frac{iH(t-t_0)}{\hbar}} |\psi(t_0)\rangle$$

Heisenberg picture :

Each quantum mechanical system is described by a constant / time independent state vector $|\Psi(t_0)\rangle$ (an element of separable hilbert space).

Each physical observable is represented by time dependent self adjoint operator which evolves in time by

$$\frac{dA_H(t)}{dt} = \frac{[A, H]}{i\hbar} + \frac{\partial A}{\partial t}$$

whose solution is

$$A(t) = e^{\frac{iH t}{\hbar}} A(0) e^{-\frac{iH t}{\hbar}}$$

Short notes (Balki sir) Lec-12,13 (contd.)

$$\langle x | \psi_i \rangle = \sqrt{\frac{2}{L}} \cos \frac{\pi x}{L}$$

$$\langle x | \psi_f \rangle = \sqrt{\frac{2}{2L}} \cos \left(\frac{\pi x}{2L} \right)$$

$$|\psi_i\rangle = I |\psi_i\rangle = \left(\int dx |x\rangle \langle x| \right) |\psi_i\rangle = \int dx \langle x | \psi_i \rangle |x\rangle$$

$$|\psi_f\rangle = I |\psi_f\rangle = \left(\int dx' |x'\rangle \langle x'| \right) |\psi_f\rangle = \int dx' \langle x' | \psi_f \rangle |x'\rangle$$

$$\begin{aligned} \langle \psi_i | \psi_f \rangle &= \sqrt{P} = \int dx \langle x | \psi_i \rangle^* \langle x | \left(\int dx' \langle x' | \psi_f \rangle |x'\rangle \right) \\ &= \int dx \psi_i^*(x) \int_{-L}^L dx' \langle x | x' \rangle \psi_f(x') \\ &= \int_{-L/2}^{+L/2} dx \psi_i^*(x) \int_{-L}^{+L} dx' \delta(x - x') \psi_f(x') \end{aligned}$$

$$\left\{P=\left|\int_{-L/2}^{+L/2}dx\,\psi_i^*(x)\psi_f(x)\right|^2\right\}$$

$$\delta(x-x')=\delta(x'-x)$$

$$\langle x|x'\rangle=\langle x'|x\rangle\;\;\hookrightarrow\;\text{Real No.}$$

$$= \int_{-L/2}^{+L/2} dx\, \psi_i^*(x) \underbrace{\left(\int_{-L}^{+L} dx' \, \delta(x'-x) \psi_f(x') \right)}_{\longrightarrow \psi_f(x)}$$

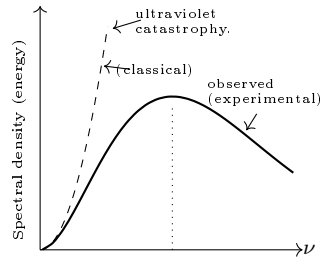
$$\boxed{\sqrt{P}=\int_{-L/2}^{L/2}dx\,\psi_i^*(x)\psi_f(x)}$$

Part II — Short Notes (Q.M), Pages 1 to 10

Short notes. (Q.M) (Page-1)

Some events / phenomena were not explained classically like ex. Colors, color vs temp, rigidity of solid, spectrum of black body radiation, discrete spectrum, stability of atom, temp. dependence of specific heat, photoelectric effect.

Plot :-



(Black body curve)

Can only be explained if radiation from oscillator is quantized. ($E = nh\nu$)

$$u_\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{\left(e^{\frac{h\nu}{kT}} - 1\right)}$$

$$u_\nu = 8\pi h \left(\frac{\nu}{c}\right)^3 \frac{1}{(e^{\beta h\nu} - 1)}$$

$$u_\nu = 8\pi h \left(\frac{\nu}{c}\right)^3 \frac{1}{(e^{\beta h\nu} - 1)}$$

u_ν = spectral density = energy per unit frequency range per unit volume.

$$\frac{dE}{d\nu} = u_\nu \Rightarrow dE = u_\nu d\nu = u_\lambda d\lambda$$

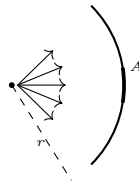
$$u_\lambda = 8\pi h \frac{c}{\lambda^5} \left(\frac{1}{e^{\frac{hc}{\lambda kT}} - 1} \right)$$

$$\frac{du}{d\lambda} = 0 \Rightarrow u_\lambda = \max \text{ at } (\lambda_{max} T = \text{const})$$

$$\Rightarrow \left(\lambda_{max} \propto \frac{1}{T} \right)$$

$$h\nu = \phi + K.E_{max} \quad \text{---(P.E) effect}$$

Intensity on a given surface :-



$$\text{No. of photons falling on area } A = \frac{N_T A}{(4\pi r^2)}$$

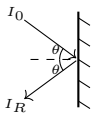
$$\text{Intensity} = \frac{\text{Energy}}{\text{Area} \times t} = \frac{N_p(A)h\nu}{A \cdot t}$$

$$I = \frac{(N_T)h\nu}{(4\pi r^2)t} \Rightarrow \left[\propto \frac{1}{r^2} \right]$$

Radiation pressure :-

(1)  $P = \frac{I}{c} \cos^2 \theta$

fully Absorbing

(2)  $P = \frac{I_0}{c} \cos^2 \theta + \frac{I_R}{c} \cos^2 \theta$

Quantization condⁿ for periodic motion

$$\text{Area in phase space} = \oint p_i dq_i = nh$$

(p, q) are generalized momenta & coordinates.

A periodic function can be expanded as

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

A non-periodic function can be expanded as

$$\begin{aligned} f(x) &= \langle x|f \rangle = \langle x| \int dk |k\rangle \langle k|f \rangle \\ &= \int \langle x|k\rangle \langle k|f \rangle dk \\ f(x) &= \int_{-\infty}^{\infty} \underbrace{e^{ikx}}_{\text{Basis}} \underbrace{f(k)}_{\text{Components}} dk \end{aligned}$$

Short notes. (Q.M) (Page-2)

Phase velocity :-

velocity of individual wave in a wave packet.

$$v_p = \frac{\omega}{k} = \frac{\Sigma \omega}{\Sigma k} = \text{speed of fast change}$$

Group velocity :-

Apparent velocity of group

$$v_g = \frac{d\omega}{dk} = \frac{\hbar d\omega}{\hbar dk} = \frac{dE}{dp}$$

$$v_g = \frac{\delta \omega}{\delta k} = \frac{|\omega_1 - \omega_2|}{|k_1 - k_2|} = \text{speed of slow change}$$

$$\Delta x \Delta p \geq \frac{\hbar}{2}; \quad \Delta E \Delta t \geq \frac{\hbar}{2}$$

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$$

$$\Delta E \updownarrow \begin{array}{c} \text{===== } E_2 \\ \\ \text{===== } E_1 \end{array}$$

{ Actual energy levels of H-atom / any atom

Virial theorem :-

Clausius introduced virial theorem in thermodynamics.

If P.E is homogeneous function of α then

$$\langle K \rangle = \frac{\alpha}{2} \langle V \rangle$$

$$\langle E \rangle = \langle T \rangle + \langle V \rangle = \left(\frac{\alpha}{2} + 1 \right) \langle V \rangle$$

LVS & vectors :-

State of Q.M. system is described by element of some LVS $|\Psi\rangle$ which satisfies

$$\|\Psi\| = \langle \Psi | \Psi \rangle = 1$$

$$\rightarrow \|\Psi\| + \|\chi\| \geq \|\Psi + \chi\|$$

$$\rightarrow \bar{a} \cdot \bar{b} \leq |a| |b|$$

$$|\langle \Psi | \chi \rangle| \leq \|\Psi\| \|\chi\|$$

$$\langle \Psi | \phi \rangle^* = \langle \phi | \Psi \rangle$$

Any element of LVS- V can be expressed as some linear combination of basis vectors.

$$|\Psi\rangle = \sum_n C_n |n\rangle$$

For continuous basis,

$$\rightarrow |\Psi\rangle = \int C_x |x\rangle$$

abstract state $|\Psi\rangle \quad C_x = \langle x|\Psi\rangle = \psi(x)$

In Q.M we use orthonormal set of basis vectors.

$$\langle \phi_i | \phi_j \rangle = \delta_{ij} \quad \text{--- orthonormality}$$

$$\sum_i |\phi_i\rangle \langle \phi_i| = I \quad \text{--- completeness.}$$

Any operator A can be expressed as

$$A = \sum_{n,m} A_{nm} |n\rangle \langle m|$$

$$(\text{or}) \quad A = \sum_i \sum_j |i\rangle A_{ij} \langle j|$$

where $\{|i\rangle\}$ forms basis for LVS.

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots \\ a_{21} & a_{22} & a_{23} & \cdots \\ \vdots & & & \end{pmatrix}$$

$$a_{ij} = A_{ij} = \langle i | \hat{A} | j \rangle$$

l_2 : Space of square summable sequences.

$L_2(a, b)$: Space of square integrable functions.

$$|\Psi\rangle = (x_1, x_2, x_3, \dots x_i)$$

$$\|\Psi\| \rightarrow \text{finite} \Rightarrow \left(\sum_{i=1}^N |x_i|^2 < \infty \right)$$

$\downarrow$

$$\text{True iff } \lim_{i \rightarrow \infty} \frac{|x_{i+1}|^2}{|x_i|^2} < 1$$

Short notes. (Q.M) (Page-3)

For $L_2(a, b)$ to be finite.

$$\text{for } \int_a^b |f(x)|^2 dx < \infty$$

$$|f(x)|^2 \rightarrow 0 \text{ faster than } |x| \rightarrow \infty$$

$$|f(x)|^2 \rightarrow 0 \text{ faster than } \frac{1}{|x|} \rightarrow 0$$

$$\text{or, } |f(x)| \rightarrow 0 \text{ faster than } \frac{1}{\sqrt{|x|}}$$

Any set of functions from (a, b) forms LVS.

$$\int_a^b f_i f_j dx = \delta_{ij}$$

We can redefine normalization conditions to define various special functions.

Generalized normalization condition.

$$\int_a^b \underset{\downarrow}{dm(x)} \phi_n^\dagger(x) \underset{\downarrow}{\phi_m(x)} = A_{nm} \delta_{nm}$$

measure weight factor

Polynomials	Range	$dm(x)$	A_{nm}
Legendre	$(-1, 1)$	1	$\frac{2}{2n+1}$
Hermite	$(-\infty, \infty)$	e^{-x^2}	?
Laguerre	$(-\infty, \infty)$	e^{-x}	?

Postulates of Q.M

(1) The state of Quantum mechanical system is described by a state vector $|\Psi(t)\rangle$ which lives in seperable-hilbert space.

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$

↓ denumerable Basis set (countable ∞ basis).

→ LVS + limit vector of cauchy sequence = complete LVS

→ complete LVS + inner products = Hilbert space

→ Hilbert space with denumerable basis is called seperable hilbert space.

(2) Every physical measurable / observable is represented by Hermitian / self adjoint operators (to ensure the result of measurement (eigen value) is always real.

$$\{\hat{x}, \hat{p}, \hat{H}, \hat{V}(x), L, S, J, \dots\}$$

All are hermitian operators.

i.e. For operator A $A^\dagger = A$

Spectrum of an operator is set of all eigen values of an operator. It can be continuous & discrete.

ex energy can be discrete & continuous



→ State of system can be written as linear combination of basis set formed by eigen vectors of observable operator.

$$|\Psi(t)\rangle = \sum_n C_n(t) |\phi_n\rangle$$

Short notes. (Q.M) (Page-4)

Operators :-

— $\langle A \rangle(t)$ in state $|\Psi(t)\rangle$

$$= \frac{\langle \Psi(t) | \hat{A} | \Psi(t) \rangle}{\langle \Psi(t) | \Psi(t) \rangle}$$

— Operators in Q.M are self adjoint or Hermitian. ($A^\dagger = A$)

— Corresponding to every transformation we use unitary operators.

$$|\Psi(t)\rangle = e^{\frac{-iHt}{\hbar}} |\Psi(0)\rangle$$

↓

Unitary operator.

Some operators

(1) $\hat{x} |x_0\rangle = x_0 |x_0\rangle$

(2) $f(\hat{x}) |x_0\rangle = f(x_0) |x_0\rangle$

(3) $\langle x | \hat{p} | \Psi \rangle = -i\hbar \frac{\partial}{\partial x} \langle x | \Psi \rangle$

(4)

Angular momentum operators :-

$$\vec{L} = \vec{r} \times \vec{p}$$

$$r = (x, y, z) \quad p = (p_x, p_y, p_z)$$

$$L_x = y p_z - z p_y$$

$$L_y = z p_x - x p_z$$

$$L_z = x p_y - y p_x$$

$$= i\hbar \begin{pmatrix} 0 & \sin \varphi & \cot \theta \cos \varphi \\ 0 & -\cos \varphi & \cot \theta \sin \varphi \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \partial_r \\ \partial_\theta \\ \partial_\varphi \end{pmatrix}$$

$$L_+ = L_x + iL_y \quad L_- = L_x - iL_y$$

Commutators :

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

(1) $[x, p] = i\hbar I$

(2) $[J_k, J_\ell] = i\hbar \epsilon_{k\ell m} J_m$

An operator is said to be hermitian if $\langle A \rangle$ is real.

$$\langle A \rangle = \langle A \rangle^*$$

$$\int \psi^\dagger A \psi dx = \int \psi (A \psi)^* dx \quad \text{---(1}\sigma\text{)}$$

$$\text{or } \int \psi^* A \psi d\tau = \int \psi (A \psi)^* d\tau \quad \text{---(3}\sigma\text{)}$$

(1) p_x is hermitian

$$\left(-i\hbar \frac{\partial}{\partial x}\right)^\dagger = -i\hbar \frac{\partial}{\partial x}$$

$$i\hbar \left(\frac{\partial}{\partial x}\right)^\dagger = -i\hbar \frac{\partial}{\partial x}$$

$$\left(\left(\frac{\partial}{\partial x}\right)^\dagger = -\frac{\partial}{\partial x}\right) \quad \left[\frac{\partial}{\partial x}\right] \text{ is not hermitian}$$

(2) $(xp_x + p_x x)$ is hermitian

$$\begin{aligned} (xp_x + p_x x)^\dagger &= p_x^\dagger x^\dagger + x^\dagger p_x^\dagger \\ &= p_x x + xp_x \end{aligned}$$

Replace xp_x from classical to $\left(\frac{xp_x + p_x x}{2}\right)$ in Q.M

$$\left(i \frac{\partial}{\partial r}\right) \text{ is Hermitian in 1-D}$$

But not in 2-D or 3-D

So always check

$$\int \psi^* A \psi d\tau = \int \psi (A \psi)^* d\tau$$

For $\hat{A}$ to be Hermitian.

Short notes. (Q.M) (Page-5)

Ehrenfest theorem :

The expectation value of operators obey classical equations.

$$\text{classically, } \frac{dA}{dt} = \{A, H\} + \frac{\partial A}{\partial t}$$

$$\text{Q.M, } \left\langle \frac{dA}{dt} \right\rangle = \frac{d}{dt} \langle A \rangle = \left\langle \frac{[A, H]}{i\hbar} \right\rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle$$

$$\Rightarrow \left\{ \frac{d}{dt} \langle p_x \rangle = \langle F_x \rangle ; \quad \frac{d \langle x \rangle}{dt} = \frac{\langle p_x \rangle}{m} \right\}$$

Unitary Operators :-

(1) space Translation: $O_T \psi(x) = \psi(x + x_0)$

$$O_T = e^{\frac{i \vec{p} \cdot \vec{r}_0}{\hbar}} \quad O_T = e^{\frac{i p_x x_0}{\hbar}}$$

(2) Time translation :-

$$\psi(t) = e^{\frac{-iHt}{\hbar}} \psi(0)$$

$$\boxed{\psi(t + t_0) = e^{\frac{-iHt_0}{\hbar}} \psi(t)}$$

Rotation about axis ($\hat{n}$)

$$O_R(\theta, \hat{n}) = e^{\frac{-i\theta(\hat{n} \cdot \vec{J})}{\hbar}}$$

Schrodinger picture :-

Operators are time independent, but wavefunctions change with time.

$$\left\{ \begin{array}{l} \hat{A} \rightarrow \text{const.} \\ \psi = \psi(t) \end{array} \right\}$$

$$\boxed{i\hbar \frac{d}{dt} |\Psi(t)\rangle = H |\Psi(t)\rangle}$$

Heisenberg picture :-

Operators are time dependent but wavefunctions are not.

$$\psi(t) = \psi(0) \text{ --- fixed}$$

$$\boxed{A_H(t) = e^{\frac{iHt}{\hbar}} A_H(0) e^{\frac{-iHt}{\hbar}}}$$

Angular momentum algebra :

- 1) $[J_i, J_j] = i\hbar \epsilon_{ijk} J_k$
- 2) $J^2 = \vec{J} \cdot \vec{J} = J_1^2 + J_2^2 + J_3^2$
- 3) $[J^2, J_i] = 0 \quad \forall i = 1, 2, 3.$
- 4) $[J^2, \vec{J} \cdot \hat{n}] = 0$ for any $\hat{n}(\theta, \varphi)$.
- 5) Since $[J^2, J_z] = 0$ they forms common set of eigen states.

$$J^2 |j, m\rangle = j(j+1)\hbar^2 |j, m\rangle$$

$$J_z |j, m\rangle = \hbar m |j, m\rangle$$

for $j, m : -j$ to $+j$ in steps of 1.

$$6) J_+ = J_x + iJ_y = a^\dagger b$$

$$J_- = J_x - iJ_y = a b^\dagger$$

$$7) J_+ |j, m\rangle = \sqrt{(j+m+1)(j-m)} |j, m+1\rangle$$

$$J_- |j, m\rangle = \sqrt{(j+m)(j-m+1)} |j, m-1\rangle$$

$$8) J_x |j, m\rangle = \frac{(J_+ + J_-)}{2} |j, m\rangle$$

$$J_y |j, m\rangle = \frac{(J_+ - J_-)}{2i} |j, m\rangle$$

Short notes. (Q.M) (Page-6)

$\langle J_i \rangle$ in given $|j, m\rangle$ state

(1)

$$\begin{aligned} \langle J_x \rangle &= \langle j, m | J_x | j, m \rangle \\ &= \langle j, m | \frac{J_+ + J_-}{2} | j, m \rangle \\ &= \langle j, m | \frac{1}{2} (a | j, m+1 \rangle \\ &\quad + b | j, m-1 \rangle) \\ &= 0 \end{aligned}$$

(2) similarly $\langle J_y \rangle$ in $|j, m\rangle = 0$

$$\langle J_y \rangle = 0$$

(3)

$$\begin{aligned} \langle J_z \rangle &= \langle j, m | J_z | j, m \rangle \\ &= \langle j, m | m\hbar | j, m \rangle \\ &= m\hbar \langle j, m | j, m \rangle \end{aligned}$$

$$\boxed{\langle J_z \rangle = m\hbar}$$

$$J_+ |j, m\rangle = \hbar \sqrt{(j+m+1)(j-m)} |j, m+1\rangle$$

$$J_- |j, m\rangle = \hbar \sqrt{(j+m)(j-m+1)} |j, m-1\rangle$$

$$\begin{aligned} [J_+, J_-] &= [J_x + iJ_y, J_x - iJ_y] \\ &= 0 - i(i\hbar J_z) + i(-i\hbar J_z) \\ &= 2\hbar J_z \end{aligned}$$

(4) Find $\langle J_x J_y \rangle$ in $|j, m\rangle$ state.

$$\begin{aligned} \langle J_x J_y \rangle &= \left\langle \left(\frac{J_+ + J_-}{2} \right) \left(\frac{J_+ - J_-}{2i} \right) \right\rangle \\ &= \left\langle \frac{(J_+)^2 - (J_-)^2 + [J_-, J_+]}{4i} \right\rangle \\ &= \frac{\langle J_+^2 \rangle}{4i} - \frac{\langle J_-^2 \rangle}{4i} + \frac{\langle [J_-, J_+] \rangle}{4i} \\ &= 0 - 0 + \left\langle \frac{2\hbar J_z}{4i} \right\rangle \\ &= \frac{\hbar i}{2} \langle j, m | J_z | j, m \rangle \end{aligned}$$

$$\langle J_x J_y \rangle = \frac{i m \hbar^2}{2}$$

(5) Find $\langle J_+ J_- \rangle$

$$\begin{aligned} \langle j, m | J_+ J_- | j, m \rangle \\ = \hbar^2 \sqrt{(j+m)(j-m+1)(j+m+1)(j-m+1)} \\ = \hbar^2 (j+m)(j-m+1) \end{aligned}$$

(6) $\langle J_- J_+ \rangle = \hbar^2 (j+m+1)(j-m)$

(7) $\langle J_x J_y + J_y J_x \rangle$

$$[J_x, J_y] = J_x J_y - J_y J_x = i \hbar J_z$$

$$\begin{aligned} \langle J_x J_y + J_y J_x \rangle &= \langle 2 J_x J_y - i \hbar J_z \rangle \\ &= 2 \langle J_x J_y \rangle - i \hbar \langle J_z \rangle \\ &= 2 \left(\frac{i m \hbar^2}{2} \right) - i \hbar m \hbar \\ &= m \hbar^2 i - m \hbar^2 i \\ &= 0 \end{aligned}$$

$$\langle J_x J_y + J_y J_x \rangle = 0$$

(8) $\langle J^2 \rangle = \langle j, m | J^2 | j, m \rangle$

$$\langle J^2 \rangle = j(j+1) \hbar^2 \langle j, m | j, m \rangle$$

$$\langle J^2 \rangle = j(j+1) \hbar^2$$

Short notes. (Q.M) (Page-7)

For $j = \frac{1}{2}$ states are represented by

$$|j, m\rangle = \begin{matrix} |\frac{1}{2}, \frac{1}{2}\rangle \\ \downarrow \end{matrix} \& \begin{matrix} |\frac{1}{2}, -\frac{1}{2}\rangle \\ \downarrow \end{matrix}$$

$$|0\rangle, |\uparrow\rangle \quad |1\rangle \text{ or } |\downarrow\rangle$$

$$|0\rangle \text{ or } |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle \text{ or } |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\bar{J}_i = \frac{\hbar}{2} \sigma_i ; \quad J^2 = J_1^2 + J_2^2 + J_3^2$$

σ_i are pauli-matrices.

Pauli matrices & their properties :-

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$|\sigma_i| = -1, \quad \sigma_i^2 = I, \quad \sigma_i \sigma_j = \delta_{ij} I + i \epsilon_{ijk} \sigma_k$$

$$[\sigma_i, \sigma_j] = 2i \epsilon_{ijk} \sigma_k$$

(4)

$$\begin{aligned} (\vec{\sigma} \cdot \vec{A})(\vec{\sigma} \cdot \vec{B}) &= (\sigma_i A_i)(\sigma_j B_j) \\ &= \sigma_i \sigma_j A_i B_j \\ &= (\delta_{ij} I + i \epsilon_{ijk} \sigma_k) A_i B_j \\ &= (\vec{A} \cdot \vec{B}) I + i \sigma_k \epsilon_{ijk} A_i B_j \\ &= (\vec{A} \cdot \vec{B}) I + i (\sigma_k)(\vec{A} \times \vec{B})_k \end{aligned}$$

$$(\vec{\sigma} \cdot \vec{A})(\vec{\sigma} \cdot \vec{B}) = (\vec{A} \cdot \vec{B}) I + i (\vec{\sigma} \cdot (\vec{A} \times \vec{B}))$$

Operators in matrix Form :-

(1)

$$\begin{aligned} J^2 &= J_1^2 + J_2^2 + J_3^2 \\ &= \frac{\hbar^2}{4} (\sigma_1^2 + \sigma_2^2 + \sigma_3^2) \end{aligned}$$

for $j = \frac{1}{2}$

$$J^2 = \frac{3\hbar^2}{4} I$$

Since $j = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$ represents spin angular momentum only,

$$\text{For } s = \frac{1}{2} \quad S^2 = \frac{3\hbar^2}{4} I$$

$$\left(\bar{S}_i = \frac{\hbar}{2} \bar{\sigma}_i \right)$$

lets write $(\vec{S} \cdot \hat{n})$ in matrix form.

$$\begin{aligned} \vec{S} \cdot \hat{n} &= \frac{\hbar}{2} \vec{\sigma} \cdot (\sin \theta \cos \varphi \hat{i} + \sin \theta \sin \varphi \hat{j} + \cos \theta \hat{k}) \\ &= \frac{\hbar}{2} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \sin \theta \cos \varphi + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \sin \theta \sin \varphi \right. \\ &\quad \left. + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cos \theta \right] \\ &= \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta \cos \varphi - i \sin \theta \sin \varphi \\ \sin \theta \cos \varphi + i \sin \theta \sin \varphi & -\cos \theta \end{pmatrix} \\ \vec{S} \cdot \hat{n} &= \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta e^{-i\varphi} \\ \sin \theta e^{i\varphi} & -\cos \theta \end{pmatrix} \end{aligned}$$

It has eigen values given by

$$|\vec{S} \cdot \hat{n} - \lambda I| = 0 \quad \boxed{\lambda = \pm \hbar/2}$$

* Eigen vectors of $\vec{S} \cdot \hat{n}$ are

$$\begin{aligned} |+\frac{\hbar}{2}\rangle &= \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i\varphi} \end{pmatrix} \\ |-\frac{\hbar}{2}\rangle &= \begin{pmatrix} \sin(\theta/2) \\ -\cos(\theta/2)e^{i\varphi} \end{pmatrix} \end{aligned}$$

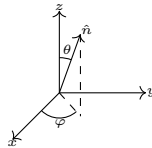
Short notes. (Q.M) (Page-7 (contd.))

$$|+\frac{\hbar}{2}\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i\varphi} \end{pmatrix} \quad |-\frac{\hbar}{2}\rangle = \begin{pmatrix} \sin(\theta/2) \\ -\cos(\theta/2)e^{i\varphi} \end{pmatrix}$$

For S_x operator :

$$\theta = 90^\circ \quad \varphi = 0$$

$$|+\frac{\hbar}{2}\rangle = \begin{pmatrix} 1/\sqrt{2} \\ +1/\sqrt{2} \end{pmatrix} ; \quad |-\frac{\hbar}{2}\rangle = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$



For S_y operator :-

$$\theta = 90^\circ \quad \varphi = 90^\circ$$

$$|+\frac{\hbar}{2}\rangle = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2}i \end{pmatrix} \quad |-\frac{\hbar}{2}\rangle = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2}i \end{pmatrix}$$

For S_z operator :

$$\theta = 0^\circ \quad \varphi = \varphi$$

$$|+\frac{\hbar}{2}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |-\frac{\hbar}{2}\rangle = \begin{pmatrix} 0 \\ -e^{i\varphi} \end{pmatrix} \text{ or } \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Probability of getting $+\hbar/2$

upon measuring S_z when system is in $|n \uparrow\rangle$ state :-

$$\begin{aligned}|n \uparrow\rangle &= \left|+\frac{\hbar}{2}(\hat{n})\right\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i\varphi} \end{pmatrix} \\ &= \cos(\theta/2) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sin \frac{\theta}{2} e^{i\varphi} \begin{pmatrix} 0 \\ 1 \end{pmatrix}\end{aligned}$$

$$P(S_z = +\frac{\hbar}{2}) = \cos^2(\theta/2)$$

$$P(S_z = -\frac{\hbar}{2}) = \sin^2(\theta/2)$$

Exponential of a matrix (2×2) :-

$$(\sigma \cdot A)(\sigma \cdot B) = I(A \cdot B) + i\vec{\sigma} \cdot (\vec{A} \times \vec{B})$$

$$(\sigma \cdot \hat{n})^2 = I$$

$$(\sigma \cdot \hat{n})^3 = \vec{\sigma} \cdot \hat{n} \Rightarrow (\sigma \cdot \hat{n})^4 = I$$

$$e^{i(\vec{\sigma} \cdot \hat{n})\theta} = I + \frac{i(\sigma \cdot \hat{n})\theta}{1!} + \frac{i^2(\sigma \cdot \hat{n})^2\theta^2}{2!} + \dots$$

$$e^{i(\vec{\sigma} \cdot \hat{n})\theta} = I(\cos \theta) + i(\vec{\sigma} \cdot \hat{n}) \sin \theta$$

$$e^{i(\vec{\sigma} \cdot \hat{n})\theta} = (\cos \theta)I + i(\vec{\sigma} \cdot \hat{n}) \sin \theta$$

To write matrix representation for J_x, J_y, J_z (for $j = j$)

Total states are set of $\{|j, m\rangle\}$

$$m : -j, -j+1, \dots, 0, \dots, j-1, j$$

Let $|1\rangle = |j, +j\rangle$

$$|2\rangle = |j, j-1\rangle \dots$$

$$|2j+1\rangle = |j, -j\rangle$$

then

$$J_i = \begin{pmatrix} \langle 1| J_i |1\rangle & \langle 1| J_i |2\rangle & \dots \\ \langle 2| J_i |1\rangle & \langle 2| J_i |2\rangle & \dots \\ \vdots & \vdots & \end{pmatrix}$$

ex. For $j = 1$ $m = -1, 0, 1$

$$|1\rangle = |1, 1\rangle \quad |2\rangle = |1, 0\rangle \quad |3\rangle = |1, -1\rangle$$

$$J_i = \begin{pmatrix} \langle 1| J_i |1\rangle & \langle 1| J_i |2\rangle & \langle 1| J_i |3\rangle \\ \langle 2| J_i |1\rangle & \langle 2| J_i |2\rangle & \langle 2| J_i |3\rangle \\ \langle 3| J_i |1\rangle & \langle 3| J_i |2\rangle & \langle 3| J_i |3\rangle \end{pmatrix}$$

$$i = 1, 2, 3.$$

Short notes. (Q.M) (Page-8)

Electron in external magnetic field :-

when an e^- is placed in ext $\vec{B}$, its intrinsic magnetic moment $\vec{\mu}_e$ interacts with $\vec{B}$.

$$\vec{\mu}_e = +\frac{gq}{2m}\vec{S}$$

$$\vec{\mu}_e = \frac{2(-|e|\hbar)}{2m_e} \frac{\hbar}{2} \vec{\sigma}$$

$$\vec{\mu}_e = \frac{-|e|\hbar}{2m_e} \vec{\sigma}$$

$$\vec{\mu}_e = -\mu_B \vec{\sigma}$$

$$\text{Interaction energy} = -\vec{m} \cdot \vec{B} = -\vec{\mu}_e \cdot \vec{B}$$

$$H = -(-\mu_B \vec{\sigma}) \cdot \vec{B}$$

$$\boxed{\hat{H} = \mu_B \vec{\sigma} \cdot \vec{B}}$$

$\hat{H}$ has two eigen values $\pm\mu_B B$

because $\vec{B} = B\hat{n}$

$$\hat{H} = \mu_B (\vec{\sigma} \cdot \hat{n}) B$$

$$E = \mu_B B(\pm 1) \quad \boxed{E = \pm\mu_B B}$$

Any particle with $(J = j)$ in $\vec{B}$.

$$\vec{\mu} = g \left(\frac{q}{2m} \right) \vec{J}$$

$$H = -\vec{\mu} \cdot \vec{B}$$

$$\hat{H} = -\frac{gq}{2m} (\vec{J} \cdot \vec{B})$$

For $\vec{B} = B\hat{n}$

$$\hat{H} = -\frac{gqB}{2m} (\vec{J} \cdot \hat{n})$$

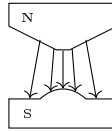
It has eigen values :

$$E = -\frac{gqB}{2m} (m\hbar) \quad (m : -j \text{ to } +j).$$

If such particles undergo force due to presence of inhomogeneous $\vec{B}$, we can find value of ' j ' by noting No. of splits.

$$F = -\nabla H$$

$$F = \frac{gq}{2m} (\vec{J} \cdot \hat{n}) \vec{\nabla} B$$

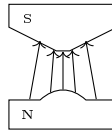


$$\vec{\nabla} B = +\hat{z}$$

$$F > 0 \text{ for } m = +j$$

$$F < 0 \text{ for } m = -j$$

If

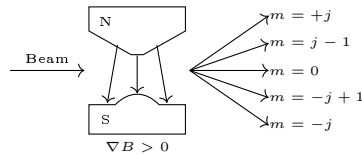


$\vec{\nabla} B < 0$ (points in dirⁿ of increasing strength).

$$F < 0 \text{ for } m = +j, j-1, \dots$$

$$F > 0 \text{ for } m = -j, -j+1, \dots$$

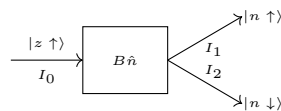
$$F = 0 \text{ for } m = 0$$



upon seeing the number of splits we can determine the value of ' j ' for the particles of Beam. This is known as Stern-Gerlach experiment.

Short notes. (Q.M) (Page-9)

General case :



Find the intensities I_1 & I_2 .

(or) $P(n \uparrow) = ?$ $P(n \downarrow) = ?$

$$|z \uparrow\rangle = a |n \uparrow\rangle + b |n \downarrow\rangle$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = a \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\varphi} \end{pmatrix} + b \begin{pmatrix} \sin \frac{\theta}{2} \\ -\cos \frac{\theta}{2} e^{i\varphi} \end{pmatrix}$$

$$P(n \uparrow) = |a|^2 = \cos^2 \frac{\theta}{2}$$

$$P(n \downarrow) = |b|^2 = \left| \begin{pmatrix} \sin \frac{\theta}{2} & -\cos \frac{\theta}{2} e^{i\varphi} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right|^2$$

$$P(n \downarrow) = \sin^2 \frac{\theta}{2}$$

$$\begin{cases} I_1 = I_0 \cos^2 \frac{\theta}{2} \\ I_2 = I_0 \sin^2 \frac{\theta}{2} \end{cases}$$

Find $\langle J_x^2 \rangle$, $\langle J_y^2 \rangle$ & $\langle J_z^2 \rangle$

$$(1) \langle J_x^2 \rangle = \langle j, m | J_x^2 | j, m \rangle$$

$$J_x = \frac{J_+ + J_-}{2}$$

$$J_x^2 = \frac{J_+^2 + J_-^2 + J_+ J_- + J_- J_+}{4}$$

$$\langle J_x^2 \rangle = \underbrace{\frac{\langle J_+^2 \rangle}{4}}_0 + \underbrace{\frac{\langle J_-^2 \rangle}{4}}_0 + \frac{\langle J_+ J_- + J_- J_+ \rangle}{4}$$

$$\begin{aligned} \langle J_+ J_- \rangle &= \langle j, m | J_+ J_- | j, m \rangle \\ &= \hbar^2 \sqrt{(j+m)(j-m+1)(j+m+1)(j-m+1)} \\ &= \hbar^2 (j+m)(j-m+1) \end{aligned}$$

Similarly, $\langle J_- J_+ \rangle = \hbar^2 (j+m+1)(j-m)$

$$\begin{aligned} \langle J_x^2 \rangle &= \frac{\hbar^2}{4} [(j+m)(j-m+1) + (j+m+1)(j-m)] \\ &= \frac{\hbar^2}{4} (j^2 - mj + j + mj - m^2 + m \\ &\quad + j^2 - jm - m^2 + j - m + mj) \\ &= \frac{\hbar^2}{4} (2j^2 + 2j - 2m^2) \end{aligned}$$

$$\boxed{\langle J_x^2 \rangle = \frac{\hbar^2}{2} (j(j+1) - m^2)}$$

$$\begin{aligned} \langle J_z^2 \rangle &= \langle j, m | J_z^2 | j, m \rangle \\ &= (m\hbar)^2 \langle j, m | j, m \rangle \\ &= m^2 \hbar^2 \end{aligned}$$

$$\begin{aligned} \langle J^2 \rangle &= \langle j, m | J^2 | j, m \rangle \\ &= \hbar^2 j(j+1) \end{aligned}$$

$$J^2 = J_x^2 + J_y^2 + J_z^2$$

$$\begin{aligned} \langle J_y^2 \rangle &= \langle J^2 \rangle - \langle J_x^2 \rangle - \langle J_z^2 \rangle \\ &= \hbar^2 j(j+1) - \frac{\hbar^2}{2} j(j+1) + \frac{m^2 \hbar^2}{2} \\ &\quad - m^2 \hbar^2 \\ &= \frac{\hbar^2 (j(j+1))}{2} - \frac{m^2 \hbar^2}{2} \end{aligned}$$

$$\boxed{\langle J_y^2 \rangle = \frac{\hbar^2}{2} (j(j+1) - m^2)}$$

So

$$\langle J_x^2 \rangle = \langle J_y^2 \rangle = \frac{\hbar^2}{2} (j(j+1) - m^2)$$

$$\langle J_z^2 \rangle = m^2 \hbar^2$$

$$\langle J^2 \rangle = \hbar^2 j(j+1)$$

Short notes. (Q.M) (Page-10)

Addition of Angular momenta and Clebsch Gordan coefficients :-

let we have two particles / or two angular momenta $\vec{J}_1$ & $\vec{J}_2$ then the resultant angular momentum is given by $\vec{J}$

$$\vec{J} = \vec{J}_1 + \vec{J}_2$$

states are represented by $|j, m\rangle$

$$J^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle$$

$$J_z |j, m\rangle = m\hbar |j, m\rangle$$

Here 'j' represents total angular momentum quantum number.

$m : -j$ to $+j$ in steps of 1

states $|j, m\rangle$ are total angular momentum states.

$$j : |j_1 - j_2| \text{ to } j_1 + j_2$$

j_1 & j_2 are quantum numbers of J_1 & J_2 respectively.

$\rightarrow (J_1^2, J_2^2, J_{1z}, J_{2z})$ commute with each other so they have common eigen states labelled by

$$|j_1 j_2, m_1 m_2\rangle$$

$$(\text{or}) \quad |j_1, m_1\rangle \otimes |j_2, m_2\rangle$$

$$(\text{or}) \quad |m_1 m_2\rangle$$

$\rightarrow (J_1^2, J_2^2, J^2, J_z)$ also commute with each other so, common eigen states are represented by

$$|j_1 j_2; j m\rangle \quad \text{or} \quad |j, m\rangle.$$

We can express any state $|j, m\rangle$ in basis of $\{|m_1 m_2\rangle\}$

$$|j, m\rangle = \sum_{m_1, m_2} C_{m_1 m_2} |m_1 m_2\rangle$$

$C_{m_1 m_2}$ are called as CB coefficients.

$$C_{m_1 m_2} = \langle m_1 m_2 | j, m \rangle.$$

To find CB coefficients :-

look for states $m = m_1 + m_2$

$$|j, m\rangle = \sum_{m_1, m_2} C_{m_1 m_2} |m_1, m_2\rangle$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ m & & (m_1 + m_2) \end{array}$$

ex $j_1 = \frac{1}{2}, j_2 = \frac{1}{2}, \quad j = |j_1 - j_2| \text{ to } j_1 + j_2$

$$m_1 = -\frac{1}{2}, \frac{1}{2} \quad m_2 = -\frac{1}{2}, \frac{1}{2} \quad j = 0 \text{ to } 1$$

we have 4 states $|0, 0\rangle, |1, -1\rangle, |1, 0\rangle, |1, 1\rangle$
 $m = 0 \quad -1, 0, 1$

we have 4 states in $|m_1 m_2\rangle$ basis.

$$|\frac{-1}{2} \frac{-1}{2}\rangle, |\frac{-1}{2} \frac{1}{2}\rangle, |\frac{1}{2} \frac{-1}{2}\rangle, |\frac{1}{2} \frac{1}{2}\rangle$$

we can see that

$$|1, 1\rangle = \begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ j & m & m_1 \quad m_2 \end{array} \begin{array}{l} |\frac{1}{2} \frac{1}{2}\rangle \\ (1 = \frac{1}{2} + \frac{1}{2}) \\ (m = m_1 + m_2) \end{array}$$

We can apply J_- to $|1, 1\rangle$ to get $|1, 0\rangle$.

$$J_- |1, 1\rangle = (J_{1-} + J_{2-}) |\frac{1}{2} \frac{1}{2}\rangle$$

also

$$\boxed{|1, -1\rangle = |\frac{-1}{2} \frac{-1}{2}\rangle}$$